WEBSCI '26 · SESSION 4B · 12 MIN

Rumors travel faster than corrections — and we can predict which ones

A structural signature of misinformation cascades, learned from 1.2M shares.

We can flag a viral rumor in its first hour — from shape alone, not content.

Corrections always lag the rumor they chase. We asked whether the *structure* of early sharing — who reshares whom, how fast, how broad — predicts a false cascade before fact-checkers weigh in. It does, with 0.81 AUC at hour one.

The problem: by the time a rumor is labelled false, it has already arrived.

On the platform we study, the median verified rumor reaches 60% of its eventual audience before any correction is posted. Content-based detectors need the text to be flagged first. We wanted a signal that fires earlier — from the cascade itself.

- Median time-to-correction: 11.4 hours.
- Median time-to-60%-reach: 2.1 hours.

Our claim: false cascades are structurally distinct from true ones, early.

True news spreads broad and shallow — many people share it once. Rumors spread deep and narrow — long chains of reshare-of-reshare. That shape difference is visible in the first hour, before the audience peaks.

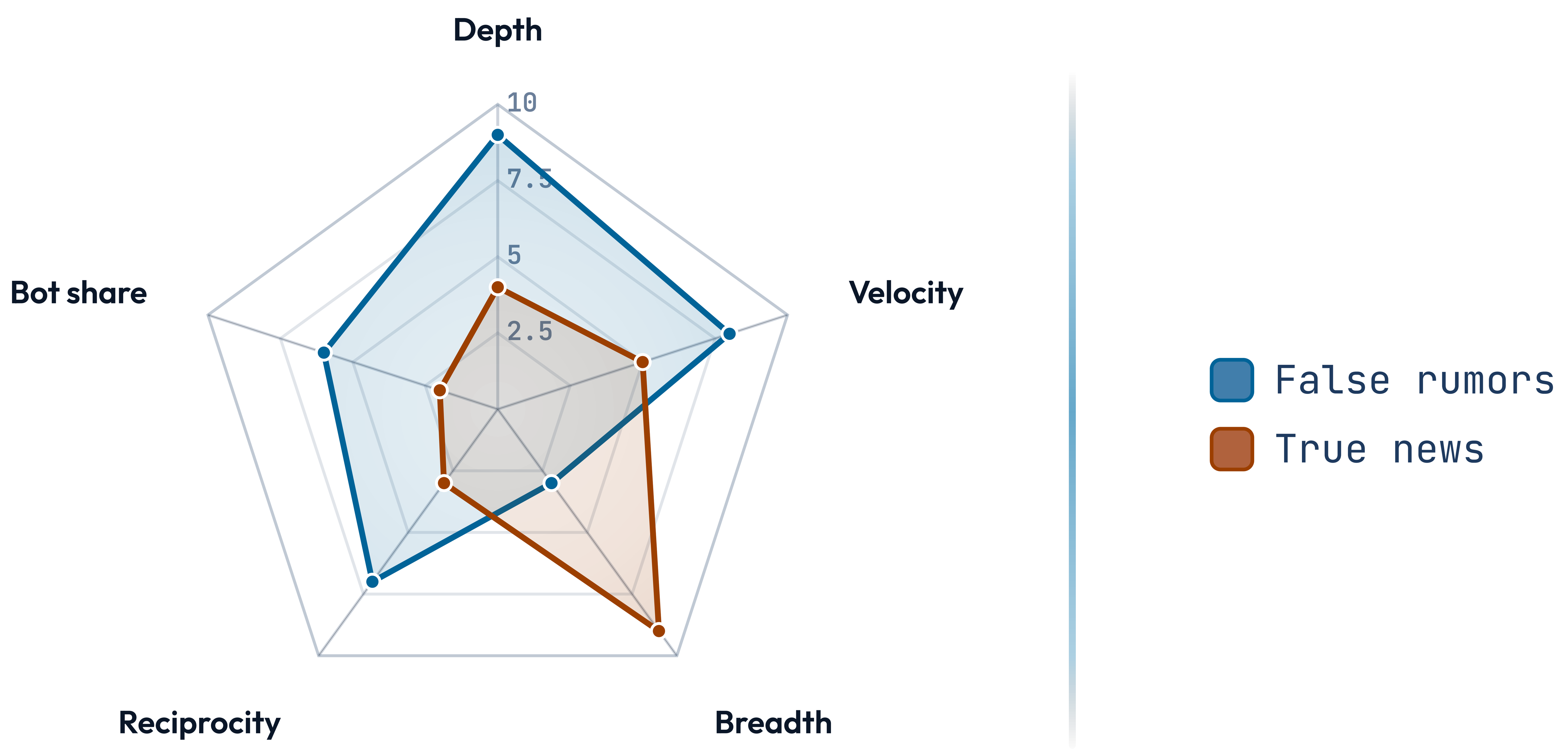
Method: 1.2M shares across 4,300 labelled cascades, modelled as growing trees.

We reconstructed each cascade as a reshare tree, sampled it at fixed time slices, and extracted twelve structural features per slice. A gradient-boosted classifier predicts the eventual fact-check verdict from the hour-one snapshot.

- Ground truth: three independent fact-check organisations, majority label.
- Strict temporal split — train on 2023–24, test on 2025.

STRUCTURAL FEATURES · 0–10 (NORMALISED)

False cascades are deeper, faster, and less broad than true ones.



HELD-OUT 2025 TEST SET · N = 1,040 CASCADES

What the structural classifier delivers at one hour.

Compared against a content-only baseline on the identical split.

0.81	+0.17	73%	1.0 h
AUC AT HOUR 1	OVER CONTENT BASELINE	PRECISION @ 50% RECALL	MEDIAN LEAD TIME GAINED

The single most predictive feature is structural virality, not volume.

Raw share count barely helps — popular true stories look popular too. What separates a rumor is *structural virality*: the average distance between any two nodes in the tree. Rumors build long chains; news builds wide fans.

- Structural virality alone reaches 0.74 AUC.
- Adding the other eleven features lifts it to 0.81.

Ablation: drop the bot-account features and performance barely moves.

We worried the model was just an automation detector. Removing every bot-related feature cost only 0.02 AUC. The structural signal is about human resharing behaviour, not coordinated inauthentic accounts — rumors spread deep because people, not bots, pass them along.

Threats to validity we want you to push on.

The labels come from English-language fact-checkers; the signature may not transfer across languages or platforms. And a model that predicts virality could be gamed by adversaries who flatten their cascades on purpose. We treat this as a triage aid, not a verdict.

- Single-platform, single-language — generalisation untested.
- Adversarial robustness is future work.

What changes if you can flag a rumor at hour one instead of hour eleven.

A ten-hour head start moves the intervention from cleanup to containment — friction prompts, reduced amplification, earlier human review. We are not proposing automated removal; we are proposing the clock start earlier.

- Pilots route hour-one flags to human moderators, not auto-action.
- The earlier the flag, the smaller the eventual audience.

DR. IMANI OSEI · CODE + DATA AT RUMORSHAPE.EXAMPLE

Rumors have a shape — and the shape shows up before the truth does.